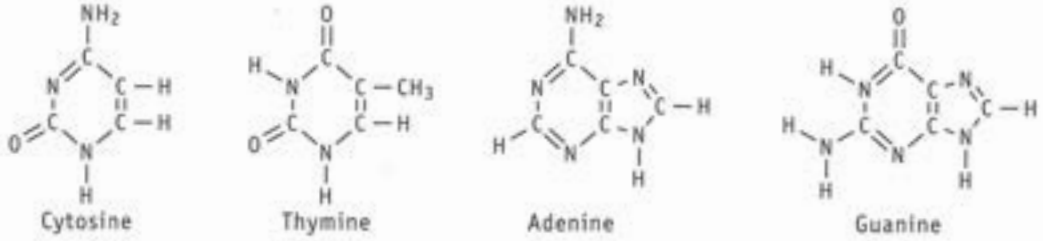
# DNA and Nucleotides

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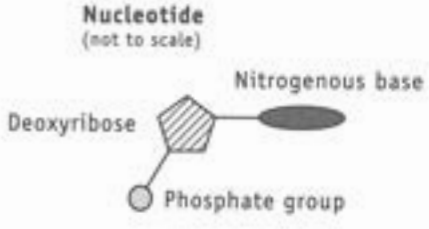
DNA can do many things, so it is amazing that it is made up of only four nitrogen-based substances (bases), sugar and phosphate molecules. The four nitrogen bases are cytosine, thymine, adenine and guanine. Cytosine and thymine are *pyrimidines*, and adenine and guanine are *purines*. Pyrimidines and purines have specific shapes.



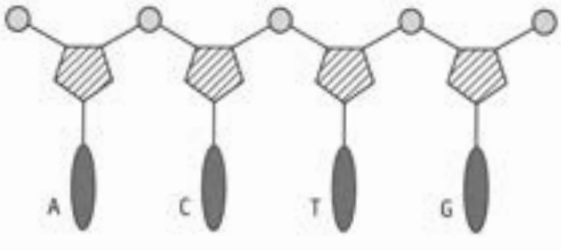

1 What elements are present in the compounds shown above?

2 What is a major difference between purines and pyrimidines?

The nitrogen base, sugar and phosphate form a three-part unit called a *nucleotide*. Each base is attached to one sugar and one phosphate molecule. The sugar involved is *deoxyribose*, hence the name *deoxyribonucleic acid* or DNA. (Deoxyribose means that the sugar is ribose, but has an oxygen atom missing). The diagram on the right shows the general structure of a nucleotide.




Study the portion of a single chain of DNA shown below. Nucleotides are held together by bonds between the sugar and phosphate groups. The nitrogen bases (adenine [A], cytosine [C], guanine [G] and thymine[T]) are side groups.




3 Copy the chain of DNA shown. (Leave plenty of space around it.)

4 Label a phosphate and a deoxyribose group on your copy of the DNA chain.

This chain will be loosely linked by hydrogen bonds to another (parallel) chain. Cytosine links with guanine, and adenine links with thymine.

5 Draw another chain of DNA parallel to and linked with the one you drew for Question 3. Identify the base used in each nucleotide.

When DNA is analysed it is found that the amounts of adenine and thymine are always equal to each other, and that the amounts of cytosine and guanine are equal to each other.

6 Explain why the amounts discussed above are equal.

The sequence of amino acids can change, by mutation for example. Mistakes can occur spontaneously — some estimates say in about one in every $10^9$ replications of base pairs. Some factors can increase the rate of mutations. High energy radiation, such as ultraviolet radiation and X-rays, can cause mutations. Some chemicals have also been suspected of causing mutations. These are called *mutagens*.

7 Write $10^9$ as a number in expanded form and in words.

8 Is mutation always harmful? Explain your answer.

9 Are hydrogen bonds strong or weak? Explain your answer.

10 What type of compound is 'ribose'?

11 What is meant by 'deoxy'?



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